MILITARY SPECIFICATION

HEMTT TANKER AVIATION REFUELING SYSTEM

(HTARS)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a refueling system tant is deployed in conjuction with the Heavy Expanded Mobility Tactical Truck (HEMTT) tanker to refuel helicopters. The system consists of dry break couplings, fittings, light weight hose, and nozzles in quantity that allows for the deployment of a four station refueling system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-F-800	- Fuel Oil, Diesel.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping Fiberboard.
PPP-T-60	- Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-G-5572	- Gasoline Aviation, Grades 80/87, 100/130, 115/145.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-N-5877	- Nozzle, Pressure Fuel Servicing, Locking, Type D-1 and D-2, Nominal 2-1/2 Inch Diameter.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-N-52747	- Nozzle Assembly, Closed Circuit Refueling.
MIL-N-52748	- Nozzle, Adapter, Closed Circuit to Gravity Fill.
MIL-H-53064	- Hose Assembly, Elastomer: Lightweight Collapsible; Liquid Fuel.
MIL-C-53071	- Coupling Assemblies, Quick-Disconnect, Sexless-Type.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type JP-8.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standard for Federal Service.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineer Design Criteria for Military Systems, Equipment and Facilities.
MS24484	- Adapter, Pressure Fuel Service, Aircraft Nominal 2-1/2 Inch Diameter.

2.1.2 Other Government drawings. The following other Government drawings, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

C13219E0479	- Adapter, Closed Circuit Fuel Servicing, Zero to 150 Gallons Per Minute.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers Inc., 345 East 47th Street New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D1.2 - Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Lejune Road, P.O. Box 351040, Miami, Fl 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The HEMTT Tanker Aviation Refueling Systems (HTARS) is a four station helicopter refueling system that consists of dry-break couplings, lighe weight hoses, and nozzles in sufficient quantity to allow for 100 foot separation between refueling points.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any deviations of HTARS from the approved first article during production, shall be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish HTARS conforming to this specification.

3.3 Materials. Materials selected by the contractor for those not specified shall be subject to all provisions of this specification. Materials shall be resistant to fluid, fluid vapors, humidity, sand, solar radiation and other environmental factors specified in this specification and shall be of the lightest practicable weight.

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3.3.1 Material deterioration prevention and control. The HTARS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the HTARS may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the HTARS may be newly fabricated from recovered materials to the maximum extent practicable, provided the HTARS produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the HTARS.

3.3.5 Fluids. The HTARS shall be compatible with MIL-F-46162, MIL-G-5572, MIL-T-5264, MIL-T-83133, and VV-F-800 fuels.

3.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for repair, replacement, an adjustment without the removal of major assemblies and installed attachements not usually removed.

3.5 Threads. Screw threads shall be in accordance with FED-STD-H28. Provisions shall be made to ensure threaded connections do not loosen due to manual testing of hoses.

3.6 Environmental.

3.6.1 Operating temperature. The HTARS shall perform as specified herein in any ambient temperature from +125 to -25 °F.

3.6.2 Storage temperature. The HTARS shall not be damaged by storage in any ambient temperature from +155 to -65 °F.

3.7 Safety. The HTARS shall not present any uncontrolled hazards to operator personnel as defined by MIL-STD-882.

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3.8 Human factors engineering. The HTARS shall conform to the human factors engineering design criteria as specified in MIL-STD-1472. Special emphasis shall be given but not limited to, General Requirements (4), Labeling (5.5), Anthropometry (5.6), Design for Maintainability (5.9), and Hazard and Safety (5.13) of MIL-STD-1472, as applicable.

3.9 HTARS components. The HTARS shall consist of the components listed in the table on figure 1. The components shall be assembled as shown in figure 1.

3.9.1 Discharge hose. The lightweight discharge hose shall be equipped with dry-break sexless couplings conforming to MIL-C-53071. The hose shall conform to MIL-H-53064. Each hose shall measure 50 feet without coupling.

3.9.1.1 Three-inch discharge hose. The 50-foot section of the three-inch discharge hose shall conform to MIL-H-53064 and be furnished with a 2-inch dry-break sexless coupling conforming to MIL-C-53071 on one end and a 3-inch female camlock type coupling on the other. The camlock shall incorporate a quarter turn valve to allow the system to be broken down without the spillage of fuel.

3.9.1.2 Hose straps. Each hose assembly shall be held in a coiled state by a nylon strap assembly. The strap shall be at least 1.5 inches wide and long enough to secure the hose when coiled and provide a 12 inch excess. The strap shall be equipped with a corrosion resistant buckle. The strap shall incorporate a loop handle. The handle shall be stitched to the main strap. The strap shall be provided with a means of securing it to the hose when not in use. The strap shall be attached to the hoseline in such a manner as to allow use of securing and releasing the buckle when the hose is coiled.

3.9.2 Nozzles. The HTARS shall be furnished with three different types of nozzles for each refueling point. A closed circuit refueling (CCR) nozzle conforming to MIL-N-52746. An open port adapter for each CCR nozzle conforming to MIL-N-52747 and a single point refueling nozzle conforming to MIL-N-5877 type D-2. The CCR and single point refueling shall connect to hose assemblies as specified in MIL-H-53064 and shall utilize pressure caps as specified in MIL-C-53071 on the inlet adapters.

3.9.3 Fittings.

3.9.3.1 Tee assemblies. The three tee assemblies shall be furnished with sexless dry-break couplings and shall conform to MIL-C-53071.

3.9.3.2 Ell assemblies. The HTARS shall be furnished with two 90-degree ell fittings conforming to MIL-C-53071 for non-drybreak sexless couplings

3.9.3.3 Recirculation nozzle. The HTARS shall be provided with a recirculation nozzle that will allow fuel to be recirculated from the HEMTT tanker through the system and reenter the truck through the bottom loading connection. The inlet side of the nozzle shall have a connection conforming to the dimensions given on drawing 13219E0479. The discharge connection shall mate with a receptacle conforming to MS24484 adapter with a misalignment angle up to 4 degrees. The nozzle shall be equipped with a port and valve to allow a fuel

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sample to be taken during recirculation. The port shall be equipped with 3 feet of clear 1/8-inch tubing that is impervious to fuel. The nozzle shall have neoprene caps on the inlet and discharge connections to protect the connections.

3.9.3.4 Emergency breakaway. Each refueling station shall be equipped with an emergency breakaway that will allow the aircraft to leave the refueling station in an emergency with the nozzle engaged. The breakaway connections shall be non-valved and conform to MIL-C-53071. The breakaway is installed inlet side of each CCR nozzle and shall seal off nozzle and fueling hose when subjected to a side pull of 200 ±50 pounds. The breakaway shall be field repairable without the use of tools. Allowable leakage during separation shall not exceed 800 cubic centimeters (cc).

3.9.3.5 Adapters. The HTARS shall be furnished with 1 2-inch sexless coupling to male camlock, valved, 1 2-inch sexless coupling to female camlock, valved, 2 2-inch sexless coupling to male camlock, nonvalve, and 2 2-inch sexless coupling to 1 1/2-inch. Fine national pipe threads. The adapters shall conform to the requirements of MIL-C-53071 as applicable.

3.10 Transportability. With all components in their transport configuration the HTARS shall fit into a box with lid with the inside dimensions 30 inches high x 50 inches long x 22 inches wide without using any specific packing scheme.

3.11 Assembly plate. Each HTARS shall be furnished with an assembly plate that pictorially depicts an installed HTARS. The plate shall contain a legend that identifies all components by associated find number. The plate shall be of material conforming to MIL-P-514, type III, composition C, grade B, class 1. The plates shall be packaged with the components of each system.

3.12 Continuity. The system shall be equipped to provide complete electrical continuity when tested in accordance with 4.6.1.

3.13 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts and without selection for fit or performance.

3.14 Workmanship.

3.14.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious t o the finished product. Manufacturing processes shall not reduce th strength of the materials to a value less than the intended design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape. External surfaces shall be free of burrs, sharp edges, and corners except where they are required or do not impact the safety of the unit.

3.14.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, grease, paint, or other foreign matter. Weldments, including weld beads,

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shall be of homogeneous apperance and form without splattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by welds. No cracks of any type are permitted. All welding is to be produced by either AWS or ASME code qualified welders. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welds shall be in accordance with AWS D1.1.

3.14.3 Welders. Before assigning any welder or welding operator to welding work, the contractor shall make available to the contracting officer certification that the welder or welding operator has passed qualification tests prescribed for the type welding to be performed and that such qualification is effective as defined by the particular code of:

AWS D1.1 - Structural Welding Code - Steel.
AWS D1.2 - Structural Welding Code - Aluminum.
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

The Government reserves the right to require recertification of any welder or welding operator.

3.14.4 Machine work. Tolerance and gauges for metal fit shall conform to commercial industry standards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.3 Acceptability criteria. HTARS which conform to all requirements in sections 3 and 5 and pass all applicable examinations and tests in section 4 of this document will be considered acceptable by the Government.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 First article examination. The first article(s) shall be examined as specified in 4.5. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article(s) shall be subjected to all tests in 4.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Each HTARS unit shall be examined as specified in 4.5. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each HTARS unit shall be subjected to the test of 4.6.1. Failure of this test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the following tests. Inspection Level S-3, AQL 6.5 shall be used for sampling:

4.6.2.1 Test conditions.

4.6.2.2 System test.

4.6.2.5 Recirculation nozzle.

4.6.2.6 Emergency breakway.

4.5 Examination. Each HTARS shall be examined as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.

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103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the HTARS.
106. Parts or components missing or not functioning as specified.
107. Assembly incorrect or not complete.
108. Threads not conforming to FED-STD-H28.
109. Discharge hose not as specified.
110. Nozzles not as specified.
111. Tee assemblies not as specified.
112. Ell assemblies not as specified.
113. Adapters not as specified.
114. Recirculation nozzles not as specified.
115. Emergency breakaway not as specified.
116. Assembly plates not as specified.
117. All like part numbers not interchangeable as specified.
118. Fabrication not as specified.
119. Welding not as specified.
120. Welders not as specified.
121. Machine work not as specified.

4.6 Tests.

4.6.1 Continuity. The HTARS shall be completely assembled and an ohmmeter used to check for a complete electrical path of less the 500,000 ohms from the origin of the system to each of the four refueling nozzles. Before connecting ohmmeter individual discharge hoses with dry-break couplings shall be subject to a 150 pound tensile load and held for 10 seconds. This load test shall be cycled 10 times. Next the hose shall be secured 3 feet from one end of the coupling. The coupling shall be rotated 360 degrees clockwise then 360 degrees counterclockwise. This shall be repeated 10 times. Failure to obtain an electrical path anywhere within the system shall constitute failue of this test.

4.6.2 Performance.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests may be conducted using clean water as a test fluid. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and fuel having a specific gravity of 0.75 at 80 °F. If water is used as a test fluid then all system components shall be immediately dried and preserved to prevent any damage or water to remain in the system.

4.6.2.2 System test. The HTARS shall be assembled for operation as specified in figure 1. The system shall be filled and pressurized to a pressure of 125 pounds per square inch (psi) for a period of 3 minutes. Any leaks, permanent deformation or failure of any component shall constitute failure of this test.

4.6.2.3 Human factors engineering. The HTARS shall be evaluated for human factors engineering characteristics throughout testing. Nonconformance to 3.8 shall constitute failure of this test.

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4.6.2.4 Safety. The system shall be evaluated for safety hazards throughout testing. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.5 Recirculation nozzle. The recirculation nozzle shall be mated with an adapter conforming to MS 24484 and a CCR nozzle conforming to 3.9.2. The test fixture shall then be pressurized to a minimum pressure of 50 psi for a period of 3 minutes. After 3 minutes, flow shall begin and continue for a period of 15 minutes at a minimum rate of 75 gallons per minute (gpm) and 25 psi. After 15 minutes the recirculation nozzle shall be disconnected and reconnected a total of 100 times from both the adapter and the CCR nozzle. After the completion of the 100 connectio the pressure/flow section of the test shall be repeated. Any leaks, permanent deformation or broken components or nonconformance to 3.9.3.3 shall constitute failure of this test.

4.6.2.6 Emergency breakaway. The emergency breakaway shall be connected to a CCR nozzle on one side and a 2 inch discharge hose on the other side. Next, apply a hydrostatic pressure of 75 psi to the breakaway for a minimum of 5 minutes. Next begin liquid flow until a condition of 50 gpm at 25 psi obtained. Then apply a side pull perpendicular to the breakaway until the breakaway separates and the nozzle is no longer attached to the breakaway. The fuel spillage shall not exceed 800 cubic centimeters. The force required to separate the breakaway shall not exceed 200 $\pm$50 pounds. Any leak, permanent deformation or breakage or nonconformance to 3.9.3.4 shall constitute failure of this test.

4.6.2.7 Transportability. All components shall be placed in a box of the size given in 3.10. The box shall contain a top and this shall be closed after all components are placed inside. The components shall be packaed using no specific packing plan. This procedure shall be repeated 5 times. Failure of all components to adequately fit into the box with the top closed or nonconformance to 3.10 shall constitute failure of this test.

4.7 Inspection of packaging.

4.7.1 First article pack inspection. The first article pack shall be examined for the defects in 4.7.2.3. The presence of any defect shall be cause for rejection of the first article pack.

4.7.2 Quality conformance inspection.

4.7.2.1 Unit of product. For purpose of inspection, a complete preserved system packed for shipment shall be considered the unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 - Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

122.	Hoses not preserved as specified for level A or C (see 5.2.1 and 5.2.2).	
123.	Nozzles not preserved as specified for level A or C (see 5.2.1.2 and 5.2.2).	

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124. Miscellaneous fittings not preserved as specified for level A or C (see 5.2.1.3 and 5.2.2).
125. Shipping containers not as specified for level A or C (see 5.3.1 and 5.3.2).
126. Strapping not zinc coated as specified for level A (see 5.3.1).
127. Marking missing, incorrect, or illegible (see 5.4).

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specificaiton. Examination shall be as specified in section 4 and shall be subject to surveillancew and approval by the Government. The first article pack may be accomplished utilizing either the first article model HTARS or a production HTARS. If the first article model HTARS is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model HTARS and production HTARS.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Discharge hoses. Discharge hoses shall be completely drained of nay test fluid and coiled flat. Dust caps shall be placed on all couplings and secured with tape or other positive means. The coiled hoses shall be secured with the straps provided.

5.2.1.2 Nozzles. Each nozzle shall be individually unit packed in a snug fitting fiberboard box conforming to PPP-B-636, class weather-resistant, type CF, grade W5c or W6c as appropriate. The boxes shall be sealed with waterproof tape conforming to PPP-T-60, type optional.

5.2.1.3 Miscellaneous fittings. Miscellaneous fittings such as adapters, tees, and elbows shall be individually preserved in accordance with method IA-8 of MIL-P-116.

5.2.2 Level C. Hoses shall be preserved as specified for level A. Nozzles shall be individually unit packed as specified for level A except the boxes shall be domestic class and closure shall be in accordance with the box specification. Miscellaneous fittings shall be preserved as specified for level A.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

5.3.1 - Level A. All system components, preserved as specified in 5.2, shall be packed in a box conforming to PPP-B-601, overseas type, style I. The box shall be the minimum size necessary to contain all items with minimum void space. Cushioning shall be used as necessary to prevent components from moving inside the shipping container. The box shall be strapped in accordance with the box specification except that strapping shall be flat steel with finish B.

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5.3.2 Level C. All components shall be packed in a snug fitting box conforming to PPP-B-636, domestic class. Cushioning shall be used as necessary to prevent movement of box contents. Closure of the box shall be in accordance with the box specification.

5.4 Marking. Unit packs and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The HTARS is intended for use with the HEMTT tanker to refuel helicopters in tactical situations. The HTARS could be exposed to all environmental conditions within the limits given within this specification. The system will be deployed over all types of terrain and hard surfaces to include sand, mud and asphalt. The system will be deployed in both daylight and darkness.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article HTARS is required (see 3.2).
c. Time frame for submission of first article pack (see 5.1).
d. Level of preservation and of packing required (see 5.2, and 5.3).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the system in accordance with this specification.

6.5 Subject term (key word) listing.

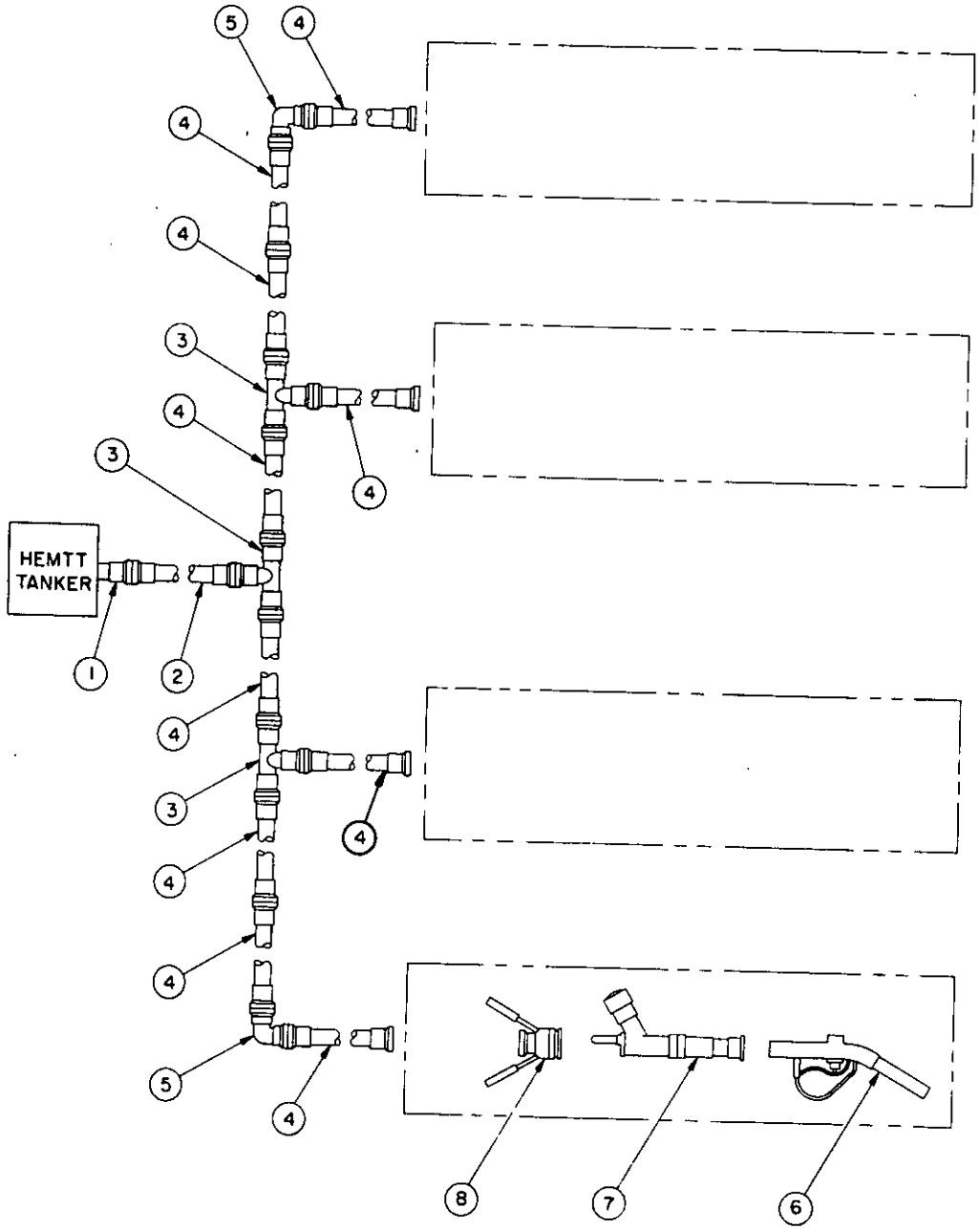
Refueling, aviation
Refueling, helicopter
Tactical refueling, helicopter

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4930-A267

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NO.	DESCRIPTION	QTY
1	ADAPTER, 3" CAM LOCK/SEXLESS	1
2	HOSE, 3" X 50ft DISCHARGE W/STRAP	1
3	TEE, 2" SEXLESS, VALVED	3
4	HOSE, 2" X 50ft DISCHARGE W/STRAP	10
5	ELBOW, 2", NO VALVE	2
6	NOZZLE, OPEN PORT	4
7	NOZZLE, CCR	4
8	NOZZLE, SINGLE PORT REFUELING	4

ADAPTERS	QTY
2" SEXLESS TO MALE CAM LOCK, VALVED	1
2" SEXLESS TO FEMALE CAM LOCK, VALVED	1
SEXLESS TO 1 1/2" F.N.P.T	2
RECIRCULATION NOZZLE	1
2" SEXLESS TO MALE CAM LOCK, NO VALVE	2

HEMTT TANKER AVIATION REFUELING SYSTEM

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-53076(ME)	HEMTT Tanker Aviation Refueling System (HTARS)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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